Ava Sadasivan

(512) 998-7074 |asadasivan7@gatech.edu|https://www.linkedin.com/in/ava-sadasivan |US Citizen

Education

Georgia Institute of Technology | Atlanta, GA, USA May 2028

Bachelor of Science in Computer Engineering, GPA 3.72/4.00

• *Relevant Coursework:* Digital Systems Design, Digital Design Lab, Circuit Analysis, Programming for HW/SW, Physics 2

Skills

Programming: Python, C, MATLAB, Swift, Arduino, RISC-V Assembly

Hardware: PCB design, soldering, wiring, embedded systems, sensor integration

Tools + Fabrication: Oscilloscope, multimeter, function generator, KiCad, Fusion 360, 3D printing, laser cutting, CNC machining

Experience

Georgia Tech — Living Dynamic Systems VIP, Atlanta, GA Jan 2025 – Apr 2025

- Designed and tested a compact EMG amplification circuit enabling real-time recording of hawkmoth flight muscle activity.
- Built analog signal amplification chain and validated signal quality during in-flight experiments

Georgia Tech — Bio-Interfaced Translational Nanoengineering Group VIP, Atlanta, GA Jan 2025- Apr 2025

- Implemented EEG preprocessing and classification pipeline based on published brain-computer interface research
- Applied Riemannian geometry feature extraction and machine learning classification to decode finger movements (89% accuracy vs 18% Euclidean baseline)

Georgia Tech Off-Road — Electrical Team Member, Atlanta, GA Sept 2025 —Dec 2025

- Wired and calibrated vehicle pressure and braking sensors for diagnostics
- Programmed Arduino microcontroller and built brake-light indicator system

Georgia Tech Hive Makerspace — Incoming Project Instructor, Atlanta, GA August 2026

- Train students on fabrication tools including CNC routers, laser cutters, and electronics equipment
- Mentor students building engineering prototypes

NASA SEES High School Internship — Robotics Team Member, Austin, TX Jun 2021 — Aug 2021

- Built autonomous rover for the ROADS on Icy Worlds Europa mission challenge
- Presented project research at the American Geophysical Union meeting

Projects

Vibrotacile Stimulation Glove for Parkinson's Patient Aug 2023 – May 2024

- Designed wearable glove using linear resonant actuators controlled via I^2C microcontroller interface
- Programmed stimulation patterns to reduce tremor amplitude
- Iterated enclosure design with foam inserts and adjustable straps for patient comfort

Make It! — STEM Discovery App May 2023 — Dec 2023

- Developed an iOS app (Swift/Xcode) recommending STEM projects based on available materials; integrated Firestore database and web-scraped tutorial dataset
- Winner — U.S. Congressional App Challenge

RISC-V Tile Matching Program — Programming for HW/SW Systems Fall 2025

- Implemented a RISC-V assembly program to identify candidate tiles matching a reference square in a grid
- Manipulated registers and memory addressing to perform pattern comparison operations